

Seeing the Invisible

Next-generation cameras leveraging hidden properties of light

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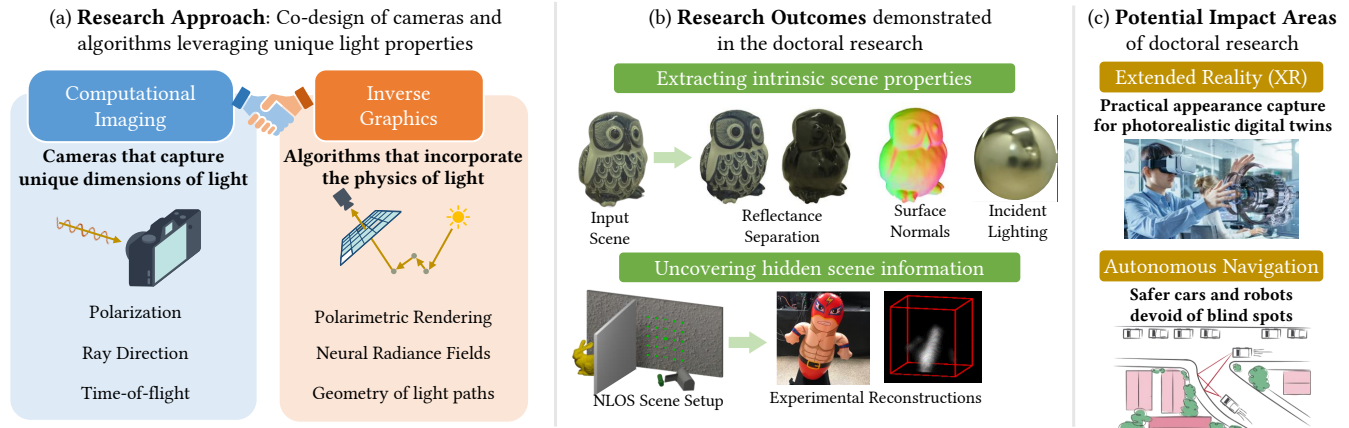


Figure 1: Overview of doctoral research: My doctoral thesis develops cameras and algorithms that can leverage dimensions of light nearly invisible to the human eye and conventional cameras (a). Through a joint design of imaging systems and physics-based inverse algorithms, my thesis demonstrates how we can extract intrinsic scene properties and uncover hidden scene information (b). My research has transformative potential in domains such as augmented reality, virtual reality, robotics, and automotive vision (c).

ABSTRACT

Conventional cameras, inspired by human vision, only capture the intensity and color of light, restricting our ability to fully comprehend the visual environment. My thesis aims to transcend the limitations of traditional RGB imaging by delving into three under-explored light properties: polarization, ray direction, and time-of-flight. Firstly, this thesis leverages polarization of surface reflections for geometry estimation, reflectance separation, and lighting estimation, resulting in accurate estimation and editing of visual appearance. Then this thesis demonstrates how to leverage reflections on multi-view images of glossy objects to recover the reflected 3D environment, essentially turning the glossy objects into cameras. Finally, this thesis develops imaging systems that exploit the distribution of time-of-flight of photons, also known as transients, to reconstruct objects hidden from the camera's line of sight. By harnessing the full potential of light, this work lays the foundation for next-generation vision systems that can recover information seemingly invisible to human eyes and conventional cameras.

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1 INTRODUCTION

The visual world is a complex interplay of light, encompassing a wealth of information beyond what we perceive with the human eye. Conventional cameras, inspired by human vision, are limited to measuring light intensity and color, thereby constraining our ability to capture the visual environment. The intensity and color of a scene are intricate functions of the scene's geometry, material properties, and lighting. Extracting these properties from RGB images alone is extremely challenging. Additionally, the intensity and color we perceive are dominated by direct light reflections. As a result, the scope of conventional cameras is limited to their direct line of sight, leading to potential blind spots. These limitations impede the potential of conventional RGB-based vision systems designed to mimic human vision and the scene information invisible to the human eye remains invisible to these systems.

But there is more to light than just intensity and color. Characteristics of a light ray such as its travel time, polarization, and direction of propagation encode meaningful scene information. The direction and polarization of light rays reveal information about the orientation and reflectance of surfaces that the ray interacted with. The time-of-flight of light can encode information about the geometry of multiple bounces of light. Harnessing these hidden dimensions

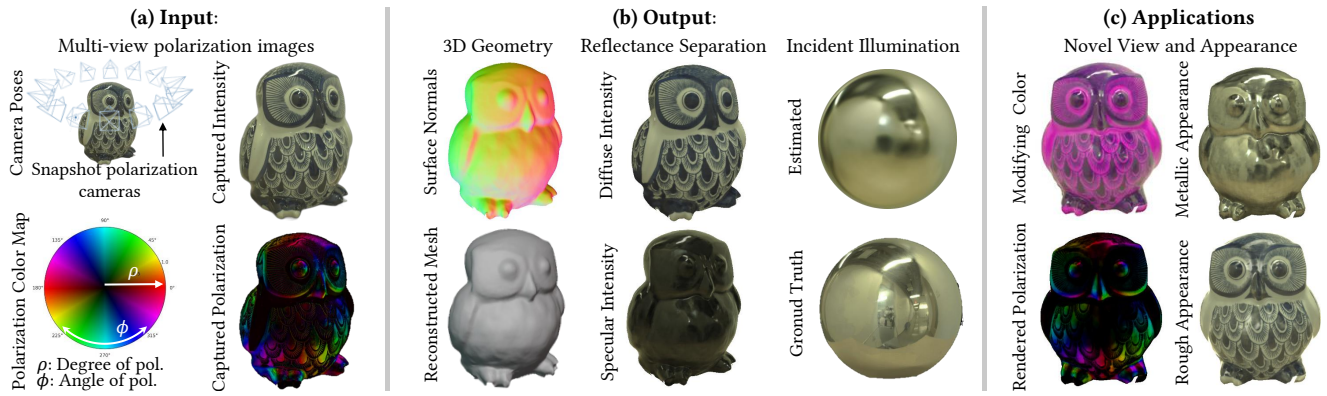


Figure 2: Contribution 1: Polarization-aided appearance decomposition We have developed computational imaging systems that demonstrate polarization is a useful and practical cue for appearance estimation [Dave et al. 2022a,b]. In PANDORA [Dave et al. 2022b], multi-view polarization images (a) enable geometry reconstruction, diffuse specular separation, and incident illumination estimation (b). From the decomposed appearance, we can synthesize the scene under novel views and novel appearance (c).

of light enables recovering information seemingly invisible in RGB measurements.

My thesis develops practical imaging systems and efficient inverse algorithms to leverage the polarization, ray directions, and time-of-flight of light (Fig. 1(a)). The developed systems demonstrate the potential to uncover seemingly hidden information such as the intrinsic surface characteristics of the objects in the scene and the shape of objects hidden from our field of view (Fig. 1(b)). The contributions of my thesis can be summarized as follows, organized by the property of light they utilize:

- **Contribution 1: Polarization for appearance decomposition.** My thesis proposes practical techniques that leverage polarization of light for 3D geometry reconstruction, reflectance separation, and incident lighting estimation (Sec. 2).
- **Contribution 2: Multi-view reflections for environment estimation** This thesis demonstrates how reflections on glossy objects can be used to image the reflected 3D environment from the object's point of view (Sec. 3).
- **Contribution 3: Time-of-flight for non line of sight imaging** This thesis shows how the distribution of photons with their time-of-flight, also called transient, encodes information about objects hidden from the camera's line of sight (Sec. 4).

The research conducted in this thesis demonstrates the remarkable potential of harnessing the time-of-flight and polarization properties of light. This work paves the way for developing next-generation vision systems that can effectively leverage emerging polarization and time-of-flight sensors. The practical computational imaging systems designed within this thesis open up transformative possibilities in several key domains (Fig. 1(c)). In AR/VR applications, extracting additional information about a scene via polarization and time-of-flight data allows for more efficient and photo-realistic, fully immersive 3D captures. In the field of autonomous navigation, these dimensions of light can be used to design safer and more collision-aware navigation systems for autonomous vehicles and robots, making them more robust in extreme weather conditions. Lastly, in medical imaging, the complex light interactions within the human body, such as the polarization sensitivity of

muscle fibers and the unique sub-surface scattering within the skin, can be better understood and visualized by capturing these additional dimensions of light, thus potentially enabling non-invasive visualization of sub-surface structures. Thus, this thesis sets the foundation for a new generation of vision systems that transcend the limitations of traditional RGB imaging.

2 POLARIZATION FOR APPEARANCE DECOMPOSITION

Polarization characterizes the direction of oscillation of light waves. The polarization of light strongly depends on the surface properties of the objects in a scene, such as surface orientation and reflectance characteristics. These surface properties define the visual appearance of an object. Thus in recent years, polarization has become a practical cue for the estimation of visual appearance [Baek et al. 2023]. Capitalizing on the emerging snapshot polarization sensors, the first contribution in my thesis establishes practical computational imaging systems for appearance estimation and decomposition. With the idea of polarization-aided appearance decomposition, two different capture modalities were explored: single-shot capture with controlled polarized lighting [Dave et al. 2022a] and multi-view capture under unstructured room lighting [Dave et al. 2022b].

2.1 Snapshot polarimetric diffuse specular separation [Dave et al. 2022a]

We present a polarization-based approach to perform diffuse-specular separation from a single polarimetric image, acquired using a flexible, practical capture setup. Our key technical insight is that we use a more general polarimetric model that accounts for partially polarized diffuse reflections instead of assuming completely unpolarized diffuse reflectance. We capture the scene with a polarimetric sensor and produce an initial analytical diffuse-specular separation that we further pass into a deep network trained to refine the separation. To our knowledge, this is the first snapshot polarimetric diffuse-specular separation approach that accounts

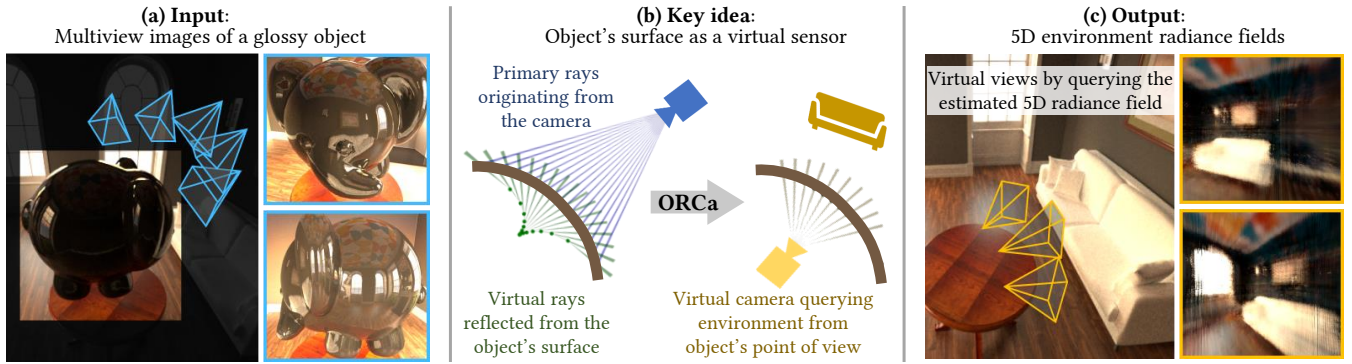


Figure 3: Multi-view reflections to turn objects into cameras We convert everyday objects with unknown geometry into radiance-field cameras by modeling multi-view reflections (a) as projections of the 5D radiance field of the environment. We convert the object surface into a virtual sensor to capture this radiance field (b), which enables depth and radiance estimation of the surrounding environment. We can then query this radiance field to perform beyond field-of-view novel view synthesis of the environment (c)

for accurate pBRDFs and works with a single uncalibrated lighting polarizer orientation. We showcase diffuse-specular separation and image-based appearance editing of real dynamic scenes. We also show that our diffuse-specular separation can be employed as a preprocessing step to enhance existing appearance acquisition.

2.2 Radiance decomposition under unstructured lighting [Dave et al. 2022b]

We propose PANDORA, a polarimetric inverse rendering approach based on implicit neural representations. From multi-view polarization images of an object, PANDORA jointly extracts the object's 3D geometry, separates the outgoing radiance into diffuse and specular components, and estimates the illumination incident on the object. Equipped with implicit surface representation and polarization cues, PANDORA outputs high-quality surface normal, signed distance field, and mesh representations of the scene. We show that PANDORA outperforms state-of-the-art radiance decomposition techniques, models strong specularities accurately, and estimates the incident illumination accurately under practical unstructured scenarios (Fig. 2).

3 MULTI-VIEW REFLECTIONS FOR ENVIRONMENT ESTIMATION

Reflections on glossy objects contain valuable and hidden information about the surrounding environment. By converting these objects into cameras, we can unlock exciting applications, including imaging beyond the camera's field of view and from seemingly impossible vantage points, e.g., from reflections on the human eye. However, this task is challenging because reflections depend jointly on object geometry, material properties, the 3D environment, and the observer viewing direction. In the second contribution of the thesis, we demonstrate how implicit neural representations enable us to extract 5D environment radiance fields from multi-view reflections on glossy objects [Tiwary and Dave et al. 2023].

3.1 Glossy objects as radiance field cameras [Tiwary and Dave et al. 2023]

We present ORCa, an approach that converts glossy objects with unknown geometry into radiance-field cameras to image the world from the object's perspective. Our key insight is to convert the object surface into a virtual sensor that captures cast reflections as a 2D projection of the 5D environment radiance field visible to the object. We show that recovering the environment radiance fields enables depth and radiance estimation from the object to its surroundings in addition to beyond field-of-view novel view synthesis, i.e. rendering of novel views that are only directly visible to the glossy object present in the scene, but not the observer. Moreover, using the radiance field we can image around occluders caused by close-by objects in the scene. Our method is trained end-to-end on multi-view images of the object and jointly estimates object geometry, diffuse radiance, and the 5D environment radiance field (Fig. 4).

4 TIME-OF-FLIGHT FOR NON LINE OF SIGHT IMAGING

The central objective of the final contribution of this thesis is to measure the distribution of the number of photons based on their respective travel times—a measure also known as the transient. By measuring transients, we can capture information from light that has taken multiple paths, including those reflected from objects hidden from the direct field of view of the camera (Fig. 4(a)). This capacity to image objects beyond the line of sight of the camera is known as non-line-of-sight (NLOS) imaging. The recent advent of single photon avalanche diode (SPAD) sensors has made capturing transients both practical and scalable. In this thesis, we exploit the geometrical properties of scattered photon light paths to conceive innovative NLOS computational imaging systems comprising pulsed lasers, SPAD detectors, and galvo mirrors [Ahn et al. 2019] (Fig. 4(b)). We have developed systems that enhance transient NLOS by borrowing ideas from two conventional imaging techniques: foveated imaging [Dave et al. 2020] and scanning-based imaging [Pediredla and Dave et al. 2019].

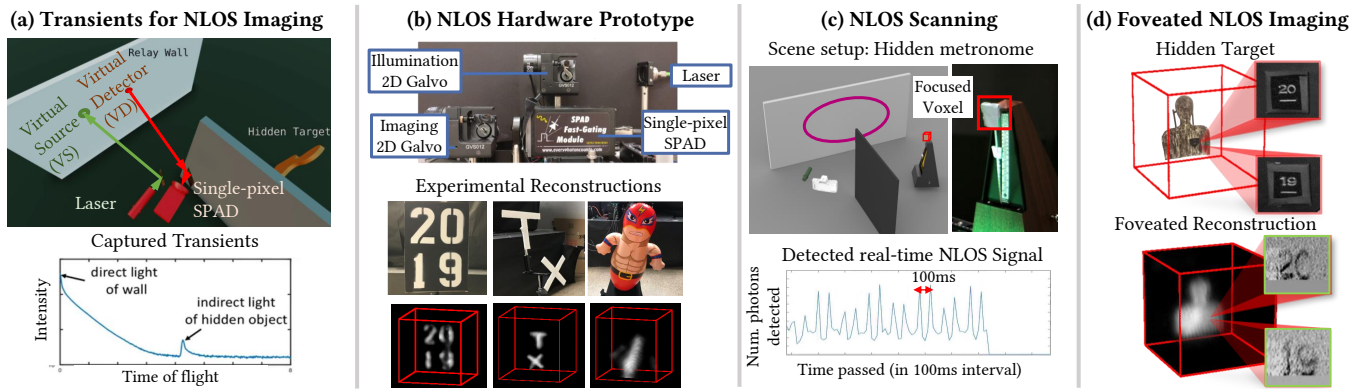


Figure 4: Transient-based Non line of sight imaging. The distribution of photons with their time of flight, also known as transients, can reveal information about objects hidden from the line of sight (a). We develop a hardware prototype comprising of pulsed laser, SPAD sensor, and galvo mirrors to demonstrate non line of sight (NLOS) imaging from transients (b). We devise a scanning-based NLOS technique to scan the hidden region voxel-by-voxel, demonstrated by recovering the real-time NLOS signal of an oscillating metronome (c). We combine information from transients with another NLOS technique: correlative, to recover coarse 3D geometry along with fine reconstruction of specular targets (d).

4.1 Foveated NLOS imaging [Dave et al. 2020]

NLOS imaging systems are inherently plagued by a tradeoff between the field of view (FOV) and spatial resolution. Transient imaging systems can reconstruct large objects at a wide FOV. Correlative based NLOS imaging systems can recover cm-scale objects at high resolution but only along a narrow FOV. We propose an imaging system that tackles this tradeoff by combining the best attributes of transient imaging and correlative subsystems. We propose an imaging pipeline that communicates information from the wide FOV sub-system to the narrow FOV sub-system in an efficient manner. Our approach provides a coarse 3D reconstruction of the hidden scene and can adaptively localize and reconstruct shiny objects in the hidden scene at higher resolution.

4.2 NLOS scanning by temporal focusing [Pediredla and Dave et al. 2019]

We present the first example of a scanning-based non-line-of-sight imaging technique. The key idea is that if the virtual sources (pulsed sources) on the wall are delayed using a quadratic delay profile (much like the quadratic phase of a focusing lens), then these pulses arrive at the same instant at a single point in the hidden volume – the point being scanned. By changing the quadratic delay profiles, we can focus light at different points in the hidden volume. This allows us to focus our measurements only on the region of interest. We derive the theoretical underpinnings of ‘temporal focusing’, show compelling simulations of performance analysis, build a hardware prototype system and demonstrate real results.

5 CONCLUSION

Our exploration into the hidden dimensions of light has revealed open challenges with polarization, multi-view reflections, and transient imaging. Polarization information can be lost due to scattering, high unpolarized background noise can interfere with polarized signals, and the use of polarizer grids can cause light attenuation.

Multi-view reflections get blurred on rough surfaces making it harder to recover the environment. Transient imaging is limited by light intensity attenuation due to scattering, the timing jitter of SPAD sensors, and the need for high data bandwidth timing circuits. Future research into more efficient and robust capture systems and inverse algorithms are crucial for overcoming these limitations.

Future research should also explore the applications of techniques developed in this thesis for characterizing volumetric scattering media. Translation of these systems to other imaging modalities, such as thermal imaging, seismic imaging, wireless, and acoustics, is another promising avenue. Continued research in leveraging these hidden properties of light would enable next-generation cameras capable of improving our understanding of the visual world.

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